

AI ASSISTED CODING

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BATCH – 03

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ASSIGNMENT – 13.2

LAB – 13 : Code Refactoring – Improving Legacy Code with AI Suggestions.

Task – 01 : Refactoring – Eliminating Repetitive Logic.

Prompt : Use AI to analyze the Python code and identify repeated calculations for area and perimeter. Refactor the code by creating a reusable function with proper docstrings while ensuring the output remains the same.

Original Code & Output :



```
TASK 01

ORIGINAL CODE

length = 8
width = 4
print("Area:", length * width)
print("Perimeter:", 2 * (length + width))

length = 12
width = 6
print("Area:", length * width)
print("Perimeter:", 2 * (length + width))

Area: 32
Perimeter: 24
Area: 72
Perimeter: 36

REFACTORED CODE
```

Refactored Code & Output :



```
REFACTORED CODE

def calculate_rectangle(length, width):
    area = length * width
    perimeter = 2 * (length + width)

    print("Area:", area)
    print("Perimeter:", perimeter)

calculate_rectangle(8, 4)
calculate_rectangle(12, 6)

Area: 32
Perimeter: 24
Area: 72
Perimeter: 36
```

Explanation :

The repeated calculations for area and perimeter were moved into a reusable function called `calculate_rectangle`. This improves code readability and reduces duplication. The program still produces the same output but is easier to maintain.

Task – 02 : Refactoring – Modularizing Inline Calculations.

Prompt : Use AI to refactor the Python code by identifying repeated discount calculations and converting them into a reusable function with proper documentation.

Original Code & Output :



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Q Commands + Code + Text Run All + RAM Disk +  
  
TASK 02  
  
ORIGINAL CODE  
  
1 amount = 1000  
2 discount = amount * 0.10  
3 final_amount = amount - discount  
4 print("Final Amount:", final_amount)  
  
5 amount = 2000  
6 discount = amount * 0.10  
7 final_amount = amount - discount  
8 print("Final Amount:", final_amount)  
  
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```

Original Code & Output :



```
Commands + Code + Text ▶ Runall
```

TASK 03

ORIGINAL CODE

```
10 students = ["Anil", "Bhavya", "Charan", "Divya"]
11 check_list = ["Charan", "Esha", "Bhavya"]
12 for name in check_list:
13     exists = False
14     for student in students:
15         if name == student:
16             exists = True
17     if exists:
18         print(name, "found")
19     else:
20         print(name, "not found")
```

→ Bhavya found

Refactored Code & Output :



```
Commands + Code + Text ▶ Runall
```

REFACTORED CODE

```
10 students = ["Anil", "Bhavya", "Charan", "Divya"]
11 check_list = ["Charan", "Esha", "Bhavya"]
12 student_set = set(students)
13
14 for name in check_list:
15     if name in student_set:
16         print(name, "found")
17     else:
18         print(name, "not found")
```

→ Charan found
→ Esha not found
→ Bhavya found

Explanation :

The nested loops were replaced with a set data structure, which provides faster lookup operations. Instead of checking each student repeatedly, membership is checked directly using `in`. This improves performance from $O(n^2)$ to $O(n)$.

Task – 04 : Refactoring – Replacing Hardcoded Values with Constants.

Prompt : Use AI to identify hardcoded numerical values in the simple interest calculation and replace them with descriptive constants to improve maintainability.

Original/Refactored Code & Output :

The screenshot shows a code editor with two panels: 'ORIGINAL CODE' and 'REFACTORED CODE'. The 'ORIGINAL CODE' panel contains the following Python code:

```
[1] print("Simple Interest:", (1000 * 2 * 5) / 100)
[2] print("Simple Interest:", (1000 * 2 * 5) / 100)
```

The 'REFACTORED CODE' panel contains the following Python code:

```
[1] RATE = 5
    TIME = 2
    PERCENT = 100
    def calculate_simple_interest(principal):
        interest = (principal * RATE * TIME) / PERCENT
        print("Simple Interest:", interest)
    calculate_simple_interest(1000)
    calculate_simple_interest(2000)
```

The output of the code is displayed below the refactored code:

```
Simple Interest: 100.0
Simple Interest: 200.0
```

Explanation :

Hardcoded values such as rate, time, and percentage were replaced with named constants (RATE, TIME, PERCENT). This makes the code easier to understand and modify. If the rate or time changes, it only needs to be updated once.

Task – 05 : Refactoring – Enhancing Variable Naming Code Clarity.

Prompt : Use AI to improve code readability by replacing unclear variable names with meaningful ones and adding comments explaining the logic.

Original/Refactored Code & Output :

The screenshot shows a code editor with two panels: 'ORIGINAL CODE' and 'REFACTORED CODE'. The 'ORIGINAL CODE' panel contains the following Python code:

```
[1] x = 50
[2] y = 30
[3] z = x + y
[4] print(z)
```

The 'REFACTORED CODE' panel contains the following Python code:

```
[1] # Define two numbers
    num1 = 50
    num2 = 30

    # Calculate the sum of the numbers
    total_sum = num1 + num2

    # Display the result
    print(total_sum)
```

The output of the code is displayed below the refactored code:

```
80
```

Explanation :

The variables x, y, and z were renamed to num1, num2, and total_sum to improve clarity. Comments were added to explain each step of the code. This makes the program easier to understand for other developers.

THANK YOU!!